A08: Independent mathematical audit of P08
Matched bilinear analysis and exact comparison

Independent reconstruction and adversarial review

3 October 2026

1 Verdicts and required correction

MB FOUNDATION VERDICT:	**VERIFIED**
MB SPECTRAL VERDICT:	**VERIFIED**
SAME-TUPLE COMPARISON VERDICT:	**MINOR REVISION**
PARITY/OBSTRUCTION VERDICT:	**VERIFIED**
OVERALL VERDICT:	**MINOR REVISION**

All four mathematical theorem groups survive independent reconstruction. The exact source nevertheless contains one false explicit example label: at the universal rational witness, the MB selected representative for $n = 8$ is 2, not 1. This is required correction E01. It does not invalidate the universal opposite-stability theorem, its strictness, the all-pair theorem, or either parity theorem. The comparison and overall verdicts conservatively retain MINOR REVISION until this source error is corrected in the original development chat and audit closure is recorded.

E01: sole required correction, P08 L534–535. Replace the two sentences in that range with:

> For $n = 5, 6, 7$ this witness selects MB representative 1; for $n = 8$ it selects representative 2. Other sizes follow the exact finite-grid rule and need not retain either label.

At $n = 8$ the representative real parts, in order $m = 0, 1, 2, 3, 4$, are

$$\left(-\frac{1}{4},\ -\frac{1}{8}+\frac{\sqrt{2}}{2},\ 1,-\frac{1}{8}-\frac{\sqrt{2}}{2},\ -\frac{9}{4}\right).$$

Their unique maximum is 1 at representative 2, giving the Fourier labels $\{2, 6\}$. In particular the erroneous label cannot be explained by conjugate-pair notation or a tie. No change to the witness tuple, equilibrium amplitudes, dispersion, or theorem statement is required.

Historical validation provenance: reproduced separately. The companion was absent from the attached 98-member archive but was located in the existing workspace as `output/p08/P08_validation_support.zip`. It was authenticated against all 98 current project member hashes and the exact input ZIP hash. After completing the independent derivations and checker, the original checker was inspected and rerun separately. Its counts were independently derived from the code's loops before comparison; all 271/11/280/28/198 counts and all numerical results reproduce exactly. This removes the provisional missing-companion limitation considered during the audit. There is no unresolved A08

provenance issue. Earlier A04/A05 historical notes are unchanged. Crucially, the historical checker did not assert the incorrect $n = 8$ label in E01: its passing run does not make that sentence true.

No P08 proof, project manifest, registry, compendium, previous audit, or later proof is edited or promoted by this audit. In particular an audit verdict is not a change to P08's current DRAFT lifecycle. No literature survey, novelty promotion, or proof after P08 is undertaken.

2 Exact current input and authentication

The authority is the supplied `cyclic_network_project_P08_draft.zip`, whose SHA-256 is

`3cac0e6676575e1c652454a9e28418c04ac5a553d4550c6f8d27f2d873e71e44`

Its current manifest entry explicitly selects `proofs/P08_matched_bilinear_analysis.tex`, **P08 v1.0, 2026-10-03**, **39,287 bytes, 846 lines**. Its full-file SHA-256 is

`d21a77de84df4c40c2995d30ab740130795adca6c72f52fef9453ad5ddfb18ac`

This agrees exactly with the manifest's final P08 record. The body hash, from the first line-anchored section command through EOF, is

`3c4393e8a8285f946a67634addbd71d617985f7f153f4963fdcbde90ca123a0d`

All P08 source ranges in this audit count comments and blank lines in that exact file, not the older placeholder or a reconstructed proof.

The pre-existing local project directory contained stale versions, including a 34-line P08 placeholder. They were excluded. A separate working extraction retained the root name `cyclic_network_project/` and all 98 archive members were authenticated against the archive bytes. The delivered audit also retains the exact manifest slug, hence its name is `audits/A08_matched_bilinear_analysis.tex`, rather than introducing a new proof slug. The audit-only ZIP contains no replacement proof file.

Before mathematical review, the current manifest, registry, frozen theorem package, final novelty verdict, novelty decisions, dependency map, canonical model, notation, and compendium source were located and consulted. Current P01–P05 and A01–A05 were located, together with P08; the shared macro file was checked for conventions. Full hashes, sizes and line counts of these 21 inputs are in `A08_support/input_hashes.json`; the complete 98-member inventory is in `A08_support/archive_member_hashes.json`.

All five prerequisite full-file hashes match the current manifest. Their body hashes match the current proof-header closure records. P01's larger-root wording, P02's ordered branch labels, and P05's corrected candidate-versus-selected-set sentence are present. The A01/A02/A05 original minor verdicts are preserved historical evidence, with the required changes closed in the current proofs. P03/P04 have no required mathematical correction. A04/A05 U01 limitations remain historical and are not claimed resolved here. P02's historical comment saying that A02 source was absent is superseded by the current manifest/compendium record and the actual restored source; it does not prevent authentication.

The frozen order is P01–P05, P08, P07, P09, P10, then P11–P14; P06 is later and conditional. The mathematical comparison fixes the same positive (a, k, c, g, h), n, q, and time variable, at each law's own specified positive synchronous equilibrium. The novelty files withhold priority clearance; this audit does not revisit that separate question.

3 Independent MB foundation

For canonical $n \geq 5$, $1 \leq q < n$, and $a, k, c, g, h > 0$, the original MB field is

$$F_i(y) = ay_i - ky_i^2 + cy_iy_{i-1} - gy_iy_{i+q} - hy_iy_{i-q} = y_iL_i(y),$$

where $L_i = a - ky_i + cy_{i-1} - gy_{i+q} - hy_{i-q}$. It is polynomial on $\mathbb{R}^n$, hence locally Lipschitz. Standard local existence, uniqueness and continuation give a unique maximal classical solution through every real initial state. For every finite time within that solution's interval,

$$y_i(t) = y_i(0) \exp\left(\int_0^t L_i(y(r))\, dr \right).$$

The exponent is finite on every compact solution interval. Thus each coordinate preserves its sign, zero coordinates remain zero, and nonnegative data remain nonnegative. This proves coordinate-face invariance even with negative other coordinates. Positivity is only on the maximal interval; no positive asymptotic lower bound or global continuation follows from this identity alone. P08 respects that scope.

On $y = s\mathbf{1}$ every component equals

$$as - ks^2 + cs^2 - gs^2 - hs^2 = as - Ms^2, \qquad M = k + g + h - c.$$

Scalar uniqueness and vector uniqueness identify the lift, and their maximal intervals coincide: extension of either supplies extension of the other. The real synchronous equilibria are 0 and a/M for $M \neq 0$, with only 0 for $M = 0$. Since $a > 0$, positive synchrony is equivalent to $M > 0$, when its sole positive amplitude is $y^* = a/M$. For $M < 0$ the other real root is negative and is not a nonnegative state.

For $s_0 > 0$, setting $z = 1/s$ gives $z' = -az + M$ and hence

$$s(t) = \frac{s_0 e^{at}}{1 + (Ms_0/a)(e^{at} - 1)}.$$

Here and in the global dynamical conclusions, time is forward from $t = 0$. When $M > 0$ this is global, bounded by $\max(s_0, a/M)$, and tends to a/M. When $M = 0$ it is exactly $s_0 e^{at}$: global, exponential and unbounded as $t \to \infty$, without finite-time blow-up. When $M = -|M| < 0$, the denominator first vanishes at

$$t_b = \frac{1}{a} \log\left(1 + \frac{a}{|M| s_0} \right) > 0.$$

The denominator is positive before t_b and the numerator is positive at it, so $s(t) \to +\infty$ as $t \uparrow t_b$. This is an actual full-system solution because of invariant synchrony. Zero data remain zero for all three signs of M; no assertion about every nonsynchronous trajectory when $M \leq 0$ is warranted or made.

3.1 Maximum estimate, ties, and exact implications

For $Y = \max_i y_i$ and $I(t) = \{i : y_i = Y\}$, a finite first-order Taylor expansion at a fixed time excludes the nonmaximal coordinates locally and yields

$$D^+Y = \max_{i \in I(t)} \dot{y}_i.$$

For every such i, $0 \leq y_{i-1}, y_{i+q}, y_{i-q} \leq Y$, giving

$$\dot{y}_i \leq aY - kY^2 + cY^2 = aY - (k - c)Y^2.$$

This proof covers multiple maximizing coordinates and every allowed offset collision; it bounds each original term and never overwrites a coincident contribution.

If $k > c$, logistic comparison gives $Y(t) \le \max(Y(0), R)$, $R = a/(k-c)$, global continuation, invariance of $[0, R]^n$, uniform attraction of bounded sets, and finite absorption into every $[0, R+\varepsilon]^n$. One may dominate every initial maximum from a bounded set by the same scalar solution starting at its supremum or at R, whichever is larger. This avoids needing a positive lower bound on that supremum. The estimate alone does not show finite absorption into the exact comparison cube.

If $k = c$, it gives $Y(t) \le Y(0)e^{at}$ and thus global forward existence, but no bound uniform over $t \ge 0$. If $k < c$, its scalar upper comparison can blow up, so this estimate alone yields neither global continuation nor failure thereof. It is not a necessary condition for boundedness. The synchronous examples do prove that $M > 0$ is necessary for *every* nonnegative trajectory to be bounded, and $M \ge 0$ is necessary for *every* such trajectory to be global. Neither converse is proved or asserted. These are exactly P08's scopes.

4 Independent Jacobian and full Fourier spectrum

Differentiate before imposing any equilibrium relation. With cyclic Kronecker deltas, the derivative is

$$\begin{aligned}\frac{\partial F_i}{\partial y_j} &= (a - 2ky_i + cy_{i-1} - gy_{i+q} - hy_{i-q})\delta_{ji} \\ &\quad + cy_i\delta_{j,i-1} - gy_i\delta_{j,i+q} - hy_i\delta_{j,i-q}.\end{aligned}$$

None of the three neighbor offsets is zero in the canonical range. At $y = s\mathbf{1}$ its diagonal is $a - 2ks + cs - gs - hs = a - (k+M)s$, and the three neighbor contributions are $cs, -gs, -hs$. With $(Pz)_i = z_{i-1}$ this gives

$$DF(s\mathbf{1}) = [a - (k+M)s]I + s(cP - gP^{-q} - hP^q).$$

Only at $s = y^* = a/M > 0$ may one substitute $a = My^*$ to obtain

$$J_{\mathrm{MB}} = y^*(-kI + cP - gP^{-q} - hP^q).$$

The verified convention is $v_j^{(m)} = n^{-1/2}e^{\mathrm{i}\theta_m j}$, $\theta_m = 2\pi m/n$. Direct application gives $P^\ell v^{(m)} = e^{-\mathrm{i}\ell\theta_m}v^{(m)}$. Therefore

$$\begin{aligned}\lambda_m^{\mathrm{MB}} &= y^*[-k + ce^{-\mathrm{i}\theta_m} - ge^{\mathrm{i}q\theta_m} - he^{-\mathrm{i}q\theta_m}], \\ \Re\lambda_m^{\mathrm{MB}} &= y^*[-k + c\cos\theta_m - (g+h)\cos(q\theta_m)], \\ \Im\lambda_m^{\mathrm{MB}} &= y^*[-c\sin\theta_m + (h-g)\sin(q\theta_m)].\end{aligned}$$

The signs of the imaginary contributions are respectively $-c, -g, +h$. The geometric sum makes the Fourier matrix unitary, so these n labels give the complete eigenvalue list with algebraic multiplicity. A unitary diagonalization proves normality and semisimplicity even if several labels give the same complex eigenvalue.

At $m = 0$ the matrix formula gives

$$\lambda_0^{\mathrm{MB}} = y^*(-k + c - g - h) = -My^* = -a.$$

Independently, $f'(s) = a - 2Ms$ for the scalar restriction, so $f'(a/M) = -a$. The tangent direction is $\mathbf{1}$, identifying exactly the uniform mode. Consequently positive MB synchrony can be linearly unstable only in a nonuniform direction. This does not classify a steady or oscillatory bifurcation, or imply nonzero frequency.

4.1 All collision cases

Contributions add in the original derivative, giving

q	J_{MB}/y^*	$\Im\lambda_m^{\mathrm{MB}}/y^*$
1	$-kI+(c-h)P-gP^{-1}$	$(-c+h-g)\sin\theta_m$
$n-1$	$-kI+(c-g)P-hP^{-1}$	$(-c+g-h)\sin\theta_m$
$n/2$	$-kI+cP-(g+h)P^{n/2}$	$-c\sin\theta_m$

The final row requires even n and uses $P^{-n/2}=P^{n/2}$; its restraint real part is $-(g+h)(-1)^m$. For even n, the Nyquist value is $y^*[-k-c-(g+h)(-1)^q]$. All zero-mode values remain $-a$. The general formula is valid without coprimality; the collision checks do not impose extra restrictions on the $q=2$ comparisons.

4.2 Exact stability and the additional global bound

Put $S=g+h$ and

$$K_{n,q}=\max_m[c\cos\theta_m-S\cos(q\theta_m)].$$

At positive synchrony $\mathrm{sa}(J_{\mathrm{MB}})=y^*(K_{n,q}-k)$. Strict stability is exactly $k>K_{n,q}$ and instability is exactly $k<K_{n,q}$. At equality every real part is nonpositive with a zero maximum. Normality gives $\|e^{tJ}\|_2=e^{t\,\mathrm{sa}(J)}$, so the linear system is Lyapunov stable but not asymptotically stable. Nonlinear boundary stability remains undecided. A nonzero spectral maximum alone is not equivalent to hyperbolicity because another real part could be zero.

For the stronger full-flow estimate, let $B=-kI+cP-gP^{-q}-hP^q$, $\delta=k-K_{n,q}>0$, and $V=\sum_i y_i$. Its symmetric part has eigenvalues $-k+c\cos\theta_m-S\cos(q\theta_m)\le-\delta$; its skew part contributes zero to $y^{\mathsf{T}}By$. Hence

$$V'=aV+y^{\mathsf{T}}By\le aV-\delta\|y\|_2^2\le aV-\frac{\delta}{n}V^2.$$

Nonnegativity bounds every coordinate by V, so this excludes finite escape and yields a bounded absorbing simplex at any radius larger than na/δ. The simplex at the exact radius is invariant and attracts bounded sets. Moreover $K_{n,q}\ge c-S$ from mode zero, so $M>0$ follows. Thus strict MB spectral stability is a sufficient global dissipativity condition for the nonnegative full flow. It is not being asserted necessary for full-flow boundedness.

The witness $(9/2,7,8,1,1)$ has $M=1$, $y^*=9/2$, $x^*=1$, $\mu=23/2>0$, and zero original residuals. For $q=2$, $K_{n,2}=\max_u(2+8u-4u^2)=6$ on every grid, uniquely at $u=1$. Thus MB is stable and dissipative despite $k=7<c=8$. Its maximum is $-9/2$. DM has $d=-13, C=4, G=1/4, H=1$, vertex $4/3$, and maximum $-39/4$ at zero. This verifies the full spectra via the formulas above, not only a selected-mode label. Conversely $k>c$ alone need not imply MB spectral stability: for example $n=8, q=2, (a,k,c,g,h)=(1,2,1,2,2)$ has $M=5>0$ and the $u=0$ MB mode has normalized real part $2>0$. This auxiliary audit example simply checks P08's scope statement.

5 Exact step-two grid selection and stability

From now on the comparison is $q = 2$. Let $u_m = \cos\theta_m$, $N = \lfloor n/2 \rfloor$ and $1 = u_0 > u_1 > \cdots > u_N$. Direct substitution gives

$$\begin{aligned} p(u) = \Re\lambda^{\mathrm{MB}}/y^* &= -k + S + cu - 2Su^2 \\ &= -k + S + \frac{c^2}{8S} - 2S\left(u - \frac{c}{4S}\right)^2. \end{aligned}$$

It is strictly concave since $S > 0$. The unconstrained vertex is $v = c/(4S) > 0$ and the constrained continuous maximizer on $[-1, 1]$ is $\min(1, v)$. Both positivity and strict concavity needed by P05's generic grid result have thus been established anew for MB.

Define $\beta_j = (u_j + u_{j+1})/2$. These midpoints strictly decrease, and the cosine addition formula gives

$$\beta_j = \cos(\pi/n)\cos((2j+1)\pi/n).$$

Exactly the midpoints $0 \le j < J$, $J = \lfloor (n+1)/4 \rfloor$, are positive. To see the count, $\cos(\pi/n) > 0$ and the other angle lies between 0 and π, so positivity is precisely $2j + 1 < n/2$. Only these switches are reachable with $v > 0$.

The nearest-cosine rule follows directly from the completed square, with closed cells

$$[\beta_0, \infty), \quad [\beta_j, \beta_{j-1}] \ (1 \le j < N), \quad (-\infty, \beta_{N-1}],$$

intersected with $v > 0$. Equality at an attainable midpoint gives only the adjacent representatives; a third or nonadjacent maximal pair would contradict strict concavity at an intermediate grid point. Putting $\chi_{\mathrm{MB}} = 4S/c = 1/v$, $\alpha_j = 1/\beta_j$ for $j < J$, the strict selections are

$$\begin{array}{c|c} 0 < \chi_{\mathrm{MB}} < \alpha_0 & 0 \\ \alpha_{j-1} < \chi_{\mathrm{MB}} < \alpha_j, 1 \le j < J & j \\ \chi_{\mathrm{MB}} > \alpha_{J-1} & J \end{array}$$

and $\chi_{\mathrm{MB}} = \alpha_j$ selects $\{j, j+1\}$. An independent subtraction confirms orientation:

$$p(u_j) - p(u_{j+1}) = c(u_j - u_{j+1})(1 - \chi_{\mathrm{MB}}\beta_j).$$

The middle row is empty at $J = 1$. The endpoint $v = 1$ itself is not a discrete tie. The exact reachable set is $0, \ldots, J$; the Nyquist singleton cannot be selected. A nonzero representative represents two conjugate labels; a tie involving zero gives three labels, another reachable tie four. These counts are not eigenvalue simplicity claims.

MB realizability is independent of DM realizability: given any $v > 0$, take $g = h = 1/2$, $c = 4v$, $k > \max(0, c-1)$ and any $a > 0$. Then $M = k + 1 - c > 0$ and the desired vertex is realized. This covers every strict cell and every positive-midpoint tie.

The four terminal grid cases are

n	J	u_J	nearest at formal $v = 0$
4ℓ	ℓ	0	ℓ
$4\ell + 1$	ℓ	$\sin(\pi/(2n))$	ℓ
$4\ell + 2$	ℓ	$\sin(\pi/n)$	$\ell, \ell + 1$
$4\ell + 3$	$\ell + 1$	$-\sin(\pi/(2n))$	$\ell + 1$

The negative reachable terminal cosine is unique to the last class. For $n = 4\ell + 2$, the formal midpoint 0 is unattainable at finite positive coefficients; the positive and negative nearest cosines differ in MB

real growth by $2y^*c\sin(\pi/n) > 0$. Dividing by y^* removes the amplitude dependence; the unnormalized gap has no uniform positive lower bound along arbitrary parameter limits.

Because $y^* > 0$, stability in a selected cell j is precisely $k > S + cu_j - 2Su_j^2$. Either representative at a tie gives the same bound. Reverse inequality is instability and equality is zero abscissa. If zero is among the maxima, $M > 0$ already gives maximum $-a < 0$. All equality boundaries and signs in P08's selection theorem pass.

6 Common equilibrium domain and structural distinctions

The original DM synchronous field, after division by a positive amplitude, is

$$B(x) = a - (k+h)x + \frac{c-g}{1+x}.$$

With $A = k + h$, $\mu = a + c - g$, $b = a - A$, and $D = b^2 + 4A\mu$, the nonzero equilibrium polynomial is $Q(x) = Ax^2 - bx - \mu$. P02's complete classification gives the common positive-root domain

$$M > 0, \qquad \mu > 0 \quad \text{or} \quad (\mu = 0, b > 0) \quad \text{or} \quad (\mu < 0, b > 0, D \geq 0).$$

It includes a boundary and a double-root locus and is not wholly open. Every positive branch must be specified when two roots occur. On $\mathcal{C}_+ = \{(a,k,c,g,h) > 0 : M > 0, \mu > 0\}$, the sole positive DM root is $x_+ = (b + \sqrt{D})/(2A)$, and it is simple. P08 correctly uses this domain for the main all-pair and unique-branch contrasts and names x_- for the distinct lower-branch construction.

At a specified positive root let $s = 1 + x$. Independent differentiation and then use of $B(x) = 0$ give

$$d = -kx - \frac{c(1+2x)}{s^2}, \quad C = \frac{c}{s}, \quad G = \frac{gx}{s^2}, \quad H = hx,$$

$$\lambda_m^{\mathrm{DM}} = d + Ce^{-\mathrm{i}\theta_m} + Ge^{2\mathrm{i}\theta_m} - He^{-2\mathrm{i}\theta_m}.$$

Thus $r_m^{\mathrm{DM}} = d + Cu_m + (G - H)(2u_m^2 - 1)$, with imaginary part $-C\sin\theta_m + (G+H)\sin(2\theta_m)$. The selection controls are

$$\chi_{\mathrm{DM}} = \frac{4(H-G)}{C} = \frac{4x(hs^2 - g)}{cs}, \qquad \chi_{\mathrm{MB}} = \frac{4(g+h)}{c}.$$

In particular

$$\frac{\chi_{\mathrm{DM}}}{\chi_{\mathrm{MB}}} = \frac{x(hs^2 - g)}{s(g+h)}.$$

There is no universal ordering of them. At fixed k, c, g, h, MB's control is independent of a; DM's can move through x. Arbitrary positive pairs can in fact be realized, as proved in the next section.

DM's curvature is $4(G - H)$: convex, linear, concave for respectively $h < g/s^2$, $h = g/s^2$, $h > g/s^2$. MB's is always $-4y^*(g+h) < 0$. In the nonconcave DM cases,

$$r(1) - r(u) = (1-u)[C + 2(G-H)(1+u)] > 0 \quad (-1 \leq u < 1),$$

so zero is uniquely dominant. The common-domain tuples $x = y = 1, c = g = 4, k = 1, a = 1 + h$, $h = 1/2, 1, 2$, have $M = \mu = 1 + h > 0$ and $G - H = 1 - h$, proving all three DM curvature signs are attainable.

DM has $F_i|_{x_i=0} = cx_{i-1}$, whereas MB's same coordinate face is invariant. P01's directed predecessor propagation makes every coordinate strictly positive for $t > 0$ from a nonzero nonnegative DM initial

state. The DM uniform eigenvalue is $xB_x = \mp x\sqrt{D}/(1+x)$ on the upper/lower distinct roots and is zero at a positive double root; MB's is always $-a$. No equality of the two amplitudes or derivatives has been assumed.

P01 supplies global nonnegative existence and bounded absorption for every canonical DM tuple. MB has the sufficient dissipative regimes already proved, but the two synchronous counterexamples prevent a whole-class claim. P08's phrase "structurally dissipative" is explicitly qualified by "over the entire canonical class," with its intended meaning supplied by these preceding mathematical statements. It does not assert that every MB tuple is unbounded.

Matching controls the node number, directed offsets, coefficient tuple and count, coefficient roles and original time variable. It does not calibrate amplitudes, effective derivatives, nonlinear effects or empirical behavior. In this $q = 2$ comparison the successor contribution has opposite derivative signs in the two laws. The fixed-source limit $cz/(1+r) \to cz$ versus $crz \to 0$ as $r \downarrow 0$ also passes. If this sign discussion is quoted for collision offsets, it must refer to the individual contribution; net coincident entries are sums. No biological equivalence or quantitative superiority is claimed.

7 Every ordered pair with both models stable

The precise quantifier is: for every fixed $n \geq 5, q = 2$ and every $(j_{\mathrm{DM}}, j_{\mathrm{MB}}) \in \{0, \ldots, J\}^2$, a nonempty open subset of physical five-dimensional $\mathcal{C}_+$ has that unique pair of representatives and both equilibria Hurwitz. This includes zero and terminal representatives in all parity classes, not only interior modes.

Choose strictly interior positive controls η, ζ in the desired DM and MB cells, using $\alpha_0/2$, the arithmetic means of adjacent α's, and $2\alpha_{J-1}$ for the two endpoints and interior cells as appropriate. Set

$$c = 4, \quad S = \zeta, \quad x = 1 + \eta/\zeta, \quad s = 1 + x, \quad h = \frac{\zeta + s\eta/x}{s^2 + 1}, \quad g = \zeta - h.$$

Certainly $h > 0$; moreover $h < \zeta$ is equivalent to $\eta < \zeta xs$, which holds since $xs > \eta/\zeta$. Thus $g, h > 0$. Choose

$$k > \max\{c + S, (c-g)/(xs) - h\}, \qquad a = (k+h)x - (c-g)/s.$$

Then $a > 0$, $M > 0$, and $\mu = x[k + h + (c-g)/s] > 0$, because $k > S > g$ and $s > 1$. Substitution in the *original* fields gives

$$F_i^{\mathrm{DM}}(x\mathbf{1}) = x[a - (k+h)x + (c-g)/s] = 0, \qquad F_i^{\mathrm{MB}}((a/M)\mathbf{1}) = (a/M)[a - M(a/M)] = 0.$$

Now $(s^2+1)h - \zeta = s\eta/x$, so $x(hs^2 - g)/s = \eta$. With $c = 4$ this proves exactly $\chi_{\mathrm{DM}} = \eta$ and $\chi_{\mathrm{MB}} = \zeta$.

All MB real parts divided by y^* are at most $-k + c + S < 0$. For every DM mode, using $u \leq 1$ and $|\cos 2\theta| \leq 1$,

$$\frac{r_m^{\mathrm{DM}}}{x} \leq -k - \frac{c}{s^2} + \frac{g}{s^2} + h < -k + S < 0.$$

The c bound follows from $-(1+2x) + su \leq -x$. The latter strict inequality is valid for every positive x, c, g, h. The large-k operation adjusts a while retaining the prescribed x, c, g, h, so neither selection control moves. This addresses the principal possible failure in the all-pair construction.

Every chosen control lies strictly inside its cell; no endpoint or parity exception is omitted. Nonempty openness in all five physical parameters follows from scalar branch regularity, proved below. Distinct nonzero representatives first become possible at $J = 2$, namely $n = 7$.

At $n = 8$ the separate rational tuple $(113/4, 9, 8, 1, 1)$ gives $M = 3$, $x = 3$, $y = 113/12$, $\mu = 141/4 > 0$. Both original divided residuals vanish:

$$113/4 - 10 \cdot 3 + 7/4 = 0, \qquad 113/4 - 3(113/12) = 0.$$

DM has $(d, C, G, H) = (-61/2, 2, 3/16, 3)$ and $\chi_{\mathrm{DM}} = 45/8 > 2\sqrt{2} = \alpha_1$, selecting 2 with maximum $-443/16$. MB has $\chi_{\mathrm{MB}} = 1$, selecting 0 with maximum $-113/4$. The full spectral formulas certify both all-mode stability assertions, including unequal own equilibrium amplitudes.

8 Universal forward stability contrast and E01

Take $(a, k, c, g, h) = (1/4, 1/8, 1, 1, 1/8)$ for any $n \geq 5, q = 2$. Here $M = \mu = 1/4 > 0$. The original synchronous residuals at amplitude one are respectively

$$\frac{1}{4} - \frac{1}{8} + \frac{1}{2} - \frac{1}{2} - \frac{1}{8} = 0, \qquad \frac{1}{4} - \frac{1}{8} + 1 - 1 - \frac{1}{8} = 0.$$

Thus $x = y = 1$ by independent calculation and uniqueness, rather than an imposed matching assumption. For DM,

$$(d, C, G, H) = (-7/8, 1/2, 1/4, 1/8), \qquad r^{\mathrm{DM}}(u) = -1 + u/2 + u^2/4.$$

Its gap to the uniform value is

$$r^{\mathrm{DM}}(1) - r^{\mathrm{DM}}(u) = (1 - u)(3 + u)/4 > 0 \quad (u < 1),$$

and $r^{\mathrm{DM}}(1) = -1/4$. This proves the exact spectral abscissa and unique representative for every size, including arbitrarily large n.

MB has $S = 9/8$, $y = 1$ and

$$p(u) = 1 + u - \frac{9}{4}u^2, \qquad v_{\mathrm{MB}} = 2/9.$$

An actual grid label $J = \lfloor (n+1)/4 \rfloor$ satisfies $-1/4 < u_J \leq 1/2$. Indeed the four parity formulas apply; the only negative case has $n \geq 7$ and $|u_J| = \sin(\pi/(2n)) < \pi/14 < 1/4$. In the $4\ell + 2$ case the largest value is $\sin(\pi/6) = 1/2$ at $n = 6$, and the other nonnegative cases are smaller. Consequently

$$p(u_J) \geq 1 - \frac{1}{4} - \frac{9}{4}\left(\frac{1}{2}\right)^2 = \frac{3}{16} > 0.$$

This is a conservative valid uniform lower bound, even though its separate worst bounds on u and u^2 do not occur simultaneously. Since $J > 0$, it establishes $\max_{m \neq 0} \Re\lambda_m^{\mathrm{MB}} \geq 3/16$ without a continuous grid approximation or finite sample as proof.

There is no representative tie for any n. A tie would require $\beta_j = 2/9$. But $4\beta_j = 2\cos(2\pi j/n) + 2\cos(2\pi(j+1)/n)$ is a sum of roots of unity and their inverses, hence an algebraic integer. It would equal the noninteger rational $8/9$, impossible: the rational root theorem says a rational root of a monic integer polynomial is an integer. This establishes strict finite-grid uniqueness. The selected representative is nonzero because mode zero equals $-a = -1/4$.

At the witness $B_x = -1/4 \neq 0$ and the MB scalar derivative is $-1/4 \neq 0$. Positivity, $M > 0$, $\mu > 0$, both stability margins, and both representative gaps are strict for each fixed n. The physical open-neighborhood claim therefore passes. The draft explicitly allows this neighborhood to depend on n; it does not prove or claim one uniform neighborhood preserving the selected representatives for all n.

The incorrect finite-size sentence. For $n = 5$ the two positive-vertex candidates are $0, 1$ and instability excludes zero. For $n = 6$, $p(1/2) = 15/16$ is maximal at 1. For $n = 7$, comparing the two reachable nonzero representatives gives

$$p(u_1) - p(u_2) = (u_1 - u_2)\left[1 - \frac{9}{4}(u_1 + u_2)\right] > 0,$$

because the rational cosine intervals verified below imply $u_1 + u_2 < 41/100 < 4/9$. Thus 1 is correct for $n = 5, 6, 7$. For $n = 8$, $u_2 = 0$ and $p(0) = 1$ exceeds every other representative value listed in E01. Its strict next-mode gap is $9/8 - \sqrt{2}/2 > 0$. P08 **L534** is therefore false for $n = 8$, even though all surrounding all-n theorem conclusions are true.

9 Reverse stability on the unique positive DM domain

For any positive DM equilibrium put $s = 1 + x$, $u = \cos\theta_m$, $w = -\cos(2\theta_m)$. Independent substitution gives

$$\frac{r_m^{\mathrm{DM}}}{x} = -k + (h - g/s^2)w - \frac{c(1 + 2x - su)}{xs^2},$$
$$\frac{r_m^{\mathrm{MB}}}{y} = -k + (g + h)w + cu.$$

Here $1 + 2x - su \geq x > 0$ for $u \leq 1$.

9.1 Necessity and equality cases

On $\mathcal{C}_+$ the uniform DM value is strictly negative. If $G - H \geq 0$, zero uniquely maximizes, proving DM stable directly. If $G - H < 0$, then $h - g/s^2 > 0$. Any nonnegative DM mode must have $w > 0$, since both $-k$ and the final term are strictly negative. When also $u \geq 0$, subtraction gives

$$\frac{r_m^{\mathrm{MB}}}{y} - \frac{r_m^{\mathrm{DM}}}{x} = g(1 + s^{-2})w + cu + \frac{c(1 + 2x - su)}{xs^2} > 0.$$

Hence even a *zero* DM mode forces a positive MB mode. For $n \equiv 0, 1, 2 \pmod 4$, every reachable DM representative has a nonnegative cosine. Applying this to a maximizing representative excludes any nonnegative DM spectral abscissa if MB is strictly stable. Ties among DM representatives cause no loophole: choose any maximizer; every reachable cosine in these classes is nonnegative. Equality $u = 0$ still gives a strictly positive gap. The case $G - H = 0$ was already handled by unique uniform selection. Thus the stronger strict implication in the draft is correct, not merely an absence of a sampled unstable point.

For $n \equiv 3 \pmod 4$, the terminal representative has negative cosine. This is the exact obstruction-proof feature, rather than a parity-table guess. The gap argument cannot force MB positivity in that mode, and the following construction realizes the exception.

9.2 Sufficiency for every size in the remaining class

Let $n \equiv 3 \pmod 4$, $n \geq 7$, $\tau = \sin(\pi/(2n)) \in (0, 1/4)$, $W = 1 - 2\tau^2 > 7/8$. Then $u_J = -\tau$, $u_{J-1} = 3\tau - 4\tau^3$, and $\beta_{J-1} = \tau W$. Choose

$$h = 1, \quad c = \tau, \quad g = \tau^2/4, \quad k = 1 - 5\tau^2/2, \quad x = 2/\tau, \quad a = (k + 1)x - (\tau - \tau^2/4)/(1 + x).$$

All coefficients are positive: $k > 27/32$, $x > 8$, and the positive large term in a exceeds 8 while the subtracted term is below $1/4$. Also $M = 2 - \tau - 9\tau^2/4 > 0$, and $\mu > 0$ because $c > g$ and $a > 0$. The original DM residual is zero by the definition of a; the MB root is its own positive $y = a/M$.

For MB, $v_{\mathrm{MB}} = \tau/(4 + \tau^2) < \tau/4 < \tau W = \beta_{J-1}$, so J is uniquely selected. Its normalized maximum is exactly

$$-k + (1 + \tau^2/4)W - \tau^2 = -\tau^2(1/4 + \tau^2/2) < 0.$$

For DM, the same label gives

$$\frac{r_J^{\mathrm{DM}}}{x} = \frac{\tau^2}{2} - \frac{gW}{s^2} - \frac{\tau(1+2x+s\tau)}{xs^2}.$$

Because $1+2x+s\tau \le (2+\tau)s$, $s > x$, and $x = 2/\tau$, the sum of the two subtractions is less than

$$\frac{g+\tau(2+\tau)}{x^2} < \frac{1/64+9/16}{4}\tau^2 = \frac{37}{256}\tau^2.$$

Thus $r_J^{\mathrm{DM}}/x > 91\tau^2/256 > 0$. Concavity holds, and

$$v_{\mathrm{DM}} = \frac{\tau}{4x(s-g/s)} < \frac{\tau}{256} < \beta_{J-1},$$

since $x > 8$ and $s - g/s > 8$. Therefore DM also uniquely selects J. Every inequality is strict for every such finite n, although some margins approach zero as n grows. Branch simplicity and finite-mode continuity imply a physical open region for each fixed size. This proves sufficiency and completes the stated iff theorem.

9.3 Exact rational seven-node witness

For $(a,k,c,g,h) = (20301/1100, 17/20, 1/2, 1/100, 1)$, $M = 34/25$, $x = 10$, $y = 20301/1496$, and $\mu > 0$. The divided original residuals are exactly

$$20301/1100 - (37/20)10 + (49/100)/11 = 0, \quad 20301/1100 - (34/25)(20301/1496) = 0.$$

The three ordered nontrivial cosines are all the roots of $8u^3 + 4u^2 - 4u - 1$; this follows by applying the seventh-root geometric sum and Chebyshev identities. Exact rational evaluation gives signs $(-,+)$ on $(31/50, 63/100)$, $(+,-)$ on $(-23/100, -11/50)$, and $(-,+)$ on $(-91/100, -9/10)$. Three disjoint sign changes exhaust a cubic's degree, so the intervals isolate the roots and cosine ordering assigns u_1, u_2, u_3.

The normalized MB real parts for representatives $1, 2, 3$ are

$$-17/20 + u_1/2 - 101u_2/100, \quad -17/20 + u_2/2 - 101u_3/100, \quad -17/20 + u_3/2 - 101u_1/100.$$

Their strict rational upper bounds are respectively $-3027/10000$, $-409/10000$, $-19262/10000$; the zero mode is $-34/25$. All seven eigenvalues therefore have negative real part. DM's coefficients are $(d, C, G, H) = (-1039/121, 1/22, 1/1210, 10)$, and

$$\frac{r_2^{\mathrm{DM}}}{10} > -\frac{17}{20} - \frac{21}{2420} - \frac{23}{22000} + \frac{12099}{12100}\frac{9}{10} = \frac{9729}{242000} > \frac{1}{25}.$$

Both vertices, $25/202$ and $55/48396$, lie below $39/200 < \beta_1$ and above the negative β_2, so both selected representatives are uniquely 2. The full imaginary parts follow by substituting these exact coefficients in the displayed full spectra; the accompanying results list all seven complex eigenvalues. No decimal cosine or numerical sign is needed for the proof.

10 Full branch-labelled domain and stronger obstruction

If $4 \mid n$, the MB mode $n/4$ has $u = 0$ and normalized real part $-k + S$. Strict MB stability forces $k > S$. At *any* positive DM equilibrium, on either allowable branch, the bound already derived gives

$$r_m^{\mathrm{DM}}/x \le -k - c/s^2 + g/s^2 + h < -k + S < 0.$$

This used only $x > 0$ and the equilibrium identity, not $\mu > 0$. It covers the positive upper root, a putative positive lower root and a putative positive double root. Indeed the latter two cannot coexist with these hypotheses because their uniform eigenvalues are positive or zero, contradicting the bound. MB still has its sole positive root a/M. This proves the stronger full-domain obstruction without an omitted branch or an equality loophole. It is a strict-stability theorem; it does not silently make a boundary assertion when $k = S$.

If $4 \nmid n$, define $\rho = \max_m[-\cos(2\theta_m)]$. It satisfies $0 < \rho < 1$. Equality $\rho = 1$ would require $4m = n \pmod{2n}$ and hence $4 \mid n$. For positivity, a nearest integer to $n/4$ gives a double angle within $2\pi/n < \pi/2$ of π for $n \geq 5$, so its cosine is negative. Let

$$e = (1-\rho)/24, \quad k = 1, \quad c = h = e, \quad g = (1+e)^3 + 2e, \quad a = (1+e)(1+2e) + e/(1+e).$$

All coefficients are positive and $M = 1 + g > 0$. Writing $A = 1 + e$, direct expansion of the original DM equation gives exactly the two roots

$$x_- = e, \qquad x_+ = e + e/(1+e)^2,$$

with $\mu = -Ae^2 - e^2/A < 0$, $b = A(x_- + x_+) > 0$ and $D = A^2(x_+ - x_-)^2 > 0$. Thus these are the distinct positive roots of the original synchronous equation, not only formal roots. The branch being compared is explicitly the lower one.

At $x_- = e$, independent calculation gives

$$\lambda_0^{\mathrm{DM}} = e^2/(1+e)^2 > 0, \qquad G - H = e\left(1 + 2e/(1+e)^2\right) > 0.$$

Hence zero uniquely dominates and DM is unstable. For MB, $S = (1+e)^3 + 3e < 1 + 10e$ since $0 < e < 1$, and every real part divided by y is bounded by

$$-1 + e + S\rho < -1 + e + (1+10e)\rho = -(1-\rho) + e(1+10\rho) < -\frac{13}{24}(1-\rho) < 0.$$

This proves full MB stability with its own amplitude $y = a/M$ and strict all-mode margin. Both DM roots are simple, so the named lower root and inequalities persist locally. At $n = 6$, $\rho = 1/2$ and $e = 1/48$, giving a fully rational instance with

$$(a, k, c, g, h) = \left(\frac{61177}{56448}, 1, \frac{1}{48}, \frac{122257}{110592}, \frac{1}{48}\right), \quad x_- = \frac{1}{48}, \quad x_+ = \frac{1}{48} + \frac{48}{2401}.$$

The exact fractions are independently checked in the support results.

Consequently on the full branch-labelled common domain reverse stability exists exactly when $4 \nmid n$. On $\mathcal{C}_+$ it exists exactly when $n \equiv 3 \pmod 4$. For $n \equiv 1, 2 \pmod 4$ there is no unique-domain reverse contrast, but there is the proved lower-branch contrast. P08 does not claim a complete upper-branch classification on the remaining part of the common domain; none is inferred here. There is no conflict between these differently scoped iff statements.

11 Remaining explicit examples and their full spectra

For any $q = 2$ witness, listing x, y, d, C, G, H and its original tuple determines every complex eigenvalue exactly through

$$\lambda_m^{\mathrm{DM}} = d + C\cos\theta_m + (G - H)\cos 2\theta_m + \mathrm{i}[-C\sin\theta_m + (G+H)\sin 2\theta_m],$$
$$\lambda_m^{\mathrm{MB}} = y[-k + c\cos\theta_m - S\cos 2\theta_m + \mathrm{i}(-c\sin\theta_m + (h-g)\sin 2\theta_m)].$$

These are full complex lists, including uniform, conjugate and Nyquist labels; imaginary parts do not determine stability by themselves. The support results additionally record their numerical evaluation for every explicit common-domain rational example.

11.1 Both unstable and different modes, only the claimed n=8 case

Take $(16351/1100, 1/2, 3/2, 1/100, 1)$, $n = 8, q = 2$. All coefficients are positive, $M = 1/100$, $\mu > 0$, $x = 10$, and $y = 16351/11$. Both original divided residuals vanish:

$$16351/1100 - 15 + (149/100)/11 = 0, \quad 16351/1100 - (1/100)(16351/11) = 0.$$

DM has $(d, C, G, H) = (-1273/242, 3/22, 1/1210, 10)$ and vertex $55/16132 < \sqrt{2}/4$. Its selected representative is uniquely 2, with

$$\mathrm{sa}(J_{\mathrm{DM}})/10 = -\frac{1}{2} - \frac{63}{2420} + \frac{12099}{12100} = \frac{2867}{6050} > 0.$$

MB's vertex is $75/202$, strictly between $\sqrt{2}/4$ and $(2 + \sqrt{2})/4$, so it uniquely selects representative 1 and

$$\mathrm{sa}(J_{\mathrm{MB}})/y = -1/2 + 3\sqrt{2}/4 > 0.$$

For extra explicit full-spectrum control, the representative MB real parts divided by y are

$$\left(-\frac{1}{100}, -\frac{1}{2} + \frac{3\sqrt{2}}{4}, \frac{51}{100}, -\frac{1}{2} - \frac{3\sqrt{2}}{4}, -\frac{301}{100}\right).$$

The positive selected value exceeds $51/100$ strictly, equivalent to $75/202 > \sqrt{2}/4$. All DM and MB imaginary parts are given by the preceding exact full-spectrum formulas. There is no tie ambiguity. The small positive M explains the large MB amplitude; it is never silently identified with $x = 10$. This proves an open example at $n = 8$, not an all-size theorem about two unstable models.

11.2 Curvature witnesses, including behavior not claimed by P08

For the three tuples $(1 + h, 1, 4, 4, h)$, $h = 1/2, 1, 2$, both amplitudes are 1, $M = \mu = 1 + h > 0$, both original residuals vanish, and $(d, C, G, H) = (-4, 2, 1, h)$. Thus for all labels

$$r_m^{\mathrm{DM}} = -4 + 2u_m + (1 - h)\cos 2\theta_m, \quad \Im\lambda_m^{\mathrm{DM}} = -2\sin\theta_m + (1 + h)\sin 2\theta_m,$$
$$r_m^{\mathrm{MB}} = -1 + 4u_m - (4 + h)\cos 2\theta_m, \quad \Im\lambda_m^{\mathrm{MB}} = -4\sin\theta_m + (h - 4)\sin 2\theta_m.$$

At $n = 8$ the DM selections are $0, 0, 1$, with maxima $-3/2, -2, -4 + \sqrt{2}$; MB selects 2 with maxima $7/2, 4, 5$. These additional checked labels simply document the examples' full behavior; P08 uses them to prove curvature realizability and does not claim these particular labels for every size.

11.3 Structural growth examples outside the common domain

The tuple $(1, 1, 3, 1, 1)$ has $M = 0$, no positive MB equilibrium, and at synchronous $s_0 = 1$ the full MB trajectory is $s = e^t$. DM's positive root is $x = 1$, with original residual zero and $(d, C, G, H) = (-13/4, 3/2, 1/4, 1)$. At $q = 2$ its dispersion has vertex $1/2$ and is strictly negative for every $u \in [-1, 1]$, since its continuous maximum is $-17/8$. This stability observation is not needed for the global boundedness contrast.

The tuple $(1, 1, 4, 1, 1)$ has $M = -1$, again no positive MB equilibrium, and $s = e^t/(2 - e^t)$ blows up at $\log 2$ from $s_0 = 1$. The DM root is $x = (\sqrt{33} - 1)/4 > 0$, since $2x^2 + x - 4 = 0$, equivalently $1 - 2x + 3/(1 + x) = 0$. Its spectral coefficients are $d = -x - 4(1 + 2x)/(1 + x)^2$, $C = 4/(1 + x)$, $G = x/(1 + x)^2$, $H = x$. The master spectrum provides every label; no MB positive-root spectrum can or should be attached. In both examples DM is globally bounded by P01, regardless of local spectral behavior. The MB origin has zero residual and derivative $aI = I$, while each stated synchronous trajectory has exactly the original full MB vector derivative. These examples are not common-root stability comparisons.

12 Openness, branch regularity, and logical scope

For any positive root, $B(x) = -Q(x)/(1+x)$. Differentiating at a root gives

$$B_x = -Q'(x)/(1+x) = \mp\sqrt{D}/(1+x_\pm).$$

On $\mathcal{C}_+$, $D = b^2 + 4A\mu > 0$, so the positive upper root is simple and smoothly determined by all five physical parameters. In the lower family $\mu < 0, b > 0, D > 0$, both roots are distinct and positive; the selected lower root has $B_x = +\sqrt{D}/(1+x_-) > 0$. The MB root a/M is smooth where $M > 0$ and its scalar derivative is $-a \neq 0$.

Every constructed strict contrast therefore continues on a locally unique positive scalar branch under arbitrary sufficiently small perturbations of $(a, k, c, g, h) \in (0, \infty)^5$. Strict parameter, existence and branch conditions remain strict. For fixed n there are only finitely many Fourier values and representative comparisons, all continuous on the branch. Their strict stability signs and strict gaps persist on an intersection of open neighborhoods. A conjugate pair remains tied, as required; uniqueness always means a unique *representative*, not a unique Fourier label. This establishes genuine physical-parameter openness rather than openness only within an equilibrium-coordinate family. No full-system invertibility or hyperbolicity is required to continue the scalar synchronous root.

This verifies the all-pair family, universal forward witness, analytic reverse family, rational $n = 7$ witness, lower-branch family, rational reverse-selection example, and the $n = 8$ two-unstable example. The open neighborhoods may depend on n. A double DM root elsewhere in the full common domain is not claimed to support such an IFT argument. Selection alone does not identify the first mode to cross along a parameter path, and no path, Hopf theorem, transversality, Lyapunov coefficient or stable periodic orbit is inferred.

The five existence categories have the exact scopes claimed: both stable with equal or unequal representatives for every fixed size; DM stable/MB unstable for every size; the reverse only in its proved branch/size domains; and both unstable with distinct representatives at the exhibited $n = 8$. There is no asserted complete partition of all physical parameter space or universal claim for every category.

13 Independent computational corroboration and limitations

The checker `A08_support/independent_check.py` was written from the original equations and this audit's independent derivations. It imports no P08 validation code or results. Before running, the distinct dimensions and expected construction counts were recorded in `expected_counts.json`. With

$$\mathcal{N} = \{5, 6, \ldots, 40, 41, 42, 43, 44, 47, 51\},$$

there are 42 dimensions. The independently derived planned counts are

$$\sum_{n\in\mathcal{N}} (J(n)+1)^2 = 2665, \qquad \sum_{n\in\mathcal{N}} (J(n)+1) = 307, \qquad \sum_{n\in\mathcal{N}} J(n) = 265.$$

These count all ordered mode pairs, strict MB cells, and positive midpoint ties respectively. There are 12 dimensions in the reverse family and 32 in the lower family; the union of distinct collision offsets $\{1, n-1\} \cup \{n/2 : n \text{ even}\}$ has 104 dimension-offset pairs. These planned counts belong to the new independent checker; reproduction of the original validation is recorded separately below.

Independent check	Completed count
Universal rational witness	42
Every ordered stable mode pair	2665
Unique-branch reverse-family sizes	12
Lower-branch reverse-family sizes	32
Strict MB selection-cell constructions	307
Positive-midpoint MB ties	265
Common-domain rational witnesses, exact Fraction residuals	9
Common-domain equilibrium cases including repeated explicit checks	2760
Original-field matrix/Fourier checks, two models per collision pair	208
Maximum inequalities, unique/tied/all-tied maxima	312
Independent random physical parameter tuples	84000
Admissible positive-branch cases from those random tuples	40436
MB-stable unique-domain random cases	15741

All 25,999 implemented assertions pass; the explicit source-label comparison separately records E01 as a false claim, rather than treating that discrepancy as a passing source assertion. The two structural growth fixtures are also independently substituted in the original fields. Exact rational arithmetic checks nine common-domain explicit witnesses, cubic isolation signs and rational spectral bounds. The 80-digit Decimal calculations use an independently coded cosine Taylor series and a stated high-precision value of π, with scaled tolerance 10^{-65}. They are high-precision corroboration, not interval certification. Largest scaled original equilibrium residual is below $6.72 \cdot 10^{-80}$.

For collision checks a separate implementation of each original vector field is differentiated by complex steps, preserving repeated neighbor contributions automatically. The resulting matrices are checked by direct action on Fourier vectors, normality and numerical eigenvalues. The maximum scaled Fourier residual is below $4.72 \cdot 10^{-15}$; the maximum scaled eigenvalue-set discrepancy is below $7.43 \cdot 10^{-15}$. Full label multiplicities are supplied by the analytic unitary diagonalization, not inferred from numerical matching.

The deterministic counterexample search uses seed 20261003 and 2,000 positive tuples per dimension, with base-10 logarithms independently uniform in $[-3, 3]$. It examines both positive DM roots when present and MB's own positive root when $M > 0$, testing the parity implications with stated floating-point margins. It finds no parity counterexample. It finds no random MB-stable lower-branch example; this sampling limitation is explicit and does not weaken the exact lower-branch construction, which is separately tested in all 32 applicable sizes. No absence of random counterexamples is used as a proof.

The audit's exact arguments, not these finite ranges, prove all-n claims, the impossibility results, openness, the blow-up formula, normality, and $\lambda_0 = -a$.

13.1 Separate reproduction of the located historical companion

The located companion has SHA-256

`81352ce048f5bd569a4d3b803f2cbd246d412aa829de43f6b6d739f2521b3958`

Its archive record identifies the exact supplied P08 ZIP and all 98 final member hashes match. The inspected checker only reads its inputs and writes its result files into the explicitly supplied output folder; it was run into a separate temporary destination.

Before reading the saved summary or comparing a rerun, the expected counts were independently

derived from its loop structure. For the equilibrium/matrix cases: two families across 28 sizes give 56, the reverse and lower families add 7 and 21, the extra rational checks add 15, all ordered pairs for sizes 5–16 add 158, and distinct collision dimension-offset cases add 14, totaling 271. Exact families count $5+3+2+1=11$; strict cells plus ties count $\sum_{n=5}^{32}(2J(n)+1)=280$; two models per collision pair give 28; the maximum checks give $9(4+5+6+7)=198$.

The rerun matches those independently expected counts. Its complete summary and all per-case JSON results equal the archived originals exactly, including the numerical maxima: matrix error $1.888608823698555\cdot10^{-14}$, direct eigenvalue error $2.1152019853906705\cdot10^{-14}$, and original-equilibrium residual $1.73735320012698\cdot10^{-16}$. The inspected source, original and reproduced summaries, reproduced results, hashes, expected counts and comparison record are preserved under `A08_support/prior_validation/`. This reproduces the historical computation but does not turn it into mathematical proof or certify statements it never checks. In particular its universal-witness cases test opposite stability and a generic selection rule, without asserting the mistaken explicit $n=8$ representative 1.

14 Source-range audit ledger

The following ledger covers every line of the exact 846-line P08 file, including headings, comments, blank lines, and the internal checklist. PASS concerns the stated scope; MINOR ISSUE identifies required E01. There is no MAJOR ISSUE or UNRESOLVED range. The literal example at E01 is false, but its local correction warrants MINOR ISSUE rather than a FALSE verdict for the valid theorem group.

Lines	Status	Independent disposition
1–13	PASS	Current DRAFT metadata, canonical models, verified prerequisites and explicit exclusions.
14–33	PASS	Same positive tuple and own equilibria; strict spectral vocabulary, full candidate labels and deferred scope.
34–45	PASS	Polynomial local flow, coordinate faces and positivity only on the maximal interval.
46–62	PASS	Scalar equilibria, forward growth and blow-up with correct signs and initial-data scope.
63–78	PASS	Locally Lipschitz field and integrating-factor identity prove sign preservation.
79–93	PASS	Synchronous lift and reciprocal solution; zero data retained and maximal intervals agree.
94–110	PASS	Maximum comparison and precise global, attraction and absorption claims.
111–122	PASS	Finite maximum handles ties; nonnegative inhibitory terms may be discarded; logistic comparison.
123–129	PASS	Necessary conditions deduced only from synchronous examples; converses not asserted.
130–150	PASS	General derivative and pre-equilibrium diagonal; coincident entries add; equilibrium used only afterward.
151–171	PASS	All phases, real and imaginary signs, full spectrum, normality and negative uniform eigenvalue.
172–184	PASS	Shift convention, unitary DFT and commuting-adjoint proofs are valid.
185–191	PASS	Independent matrix and scalar derivations of the uniform eigenvalue.

Lines	Status	Independent disposition
192–211	PASS	All collision formulas, Nyquist value, conjugacy and multiplicity qualifications.
212–228	PASS	Strict and equality stability cases; hyperbolicity distinguished; additional sufficient global bound.
229–246	PASS	Symmetric-part quadratic estimate, sum logistic bound and automatic positive MB root.
247–255	PASS	Exact dissipative example with k<c; own roots and residuals; non-equivalence of sufficient bounds.
256–269	PASS	Step-two quadratic, completion of square, strict concavity and constrained versus unconstrained vertex.
270–280	PASS	Correct generic grid definitions; MB hypotheses are established before using P05.
281–297	PASS	All strict cells, closed endpoints, positive-midpoint ties and reachable representatives.
298–311	PASS	Label counts and exact selected-mode stability including uniform maximal ties.
312–326	PASS	Independent gap sign and MB-specific positive-parameter realizability.
327–343	PASS	Four parity classes, unattainable zero-vertex tie and amplitude-dependent gap qualification.
344–366	PASS	Complete common domain, unique upper root on C-plus, and explicit branch-labelled use elsewhere.
367–382	PASS	Correct imported DM coefficients and same-tuple selection controls in the q=2 comparison.
383–404	PASS	Curvature, coordinate faces, uniform derivatives and whole-class dissipativity contrast with domains qualified.
405–414	PASS	Three positive curvature witnesses and original residuals; no inference of MB boundedness from existence.
415–427	PASS	Matching limits and fixed-source limit; derivative-sign statement read in the established q=2 scope.
428–437	PASS	Every ordered pair quantified at each fixed size, with strict stable physical open sets.
438–455	PASS	Strict control choices, positive-state construction and the large-k bound.
456–466	PASS	Positivity, common unique-branch domain, exact original residuals and independent selection ratios.
467–478	PASS	Both all-mode stability bounds; adjusting a with k preserves the chosen equilibrium and controls.
479–489	PASS	Rational reverse-selection example, distinct amplitudes, modes 2/0 and both exact negative maxima.
490–503	PASS	Universal same-tuple theorem; fixed-size open regions and accidental amplitude equality stated correctly.
504–515	PASS	Exact original residuals, coefficients and both dispersions at the universal witness.
516–524	PASS	Actual finite label J and all-size positive lower bound 3/16; nonuniformity follows from -a.
525–533	PASS	Algebraic-integer exclusion of all representative ties and valid fixed-size openness.

Lines	Status	Independent disposition
534–535	MINOR ISSUE	E01: n=8 selects MB representative 2 (labels 2 and 6), not 1; n=5,6,7 select 1.
536–538	PASS	Finite Fourier label, not a continuous maximum alone, establishes instability.
539–549	PASS	Normalized modal comparison and strictly positive last numerator.
550–557	PASS	Exact unique-domain reverse-stability iff theorem and strict implication in other parity classes.
558–572	PASS	Uniform branch stability, forced positive w and strict modal gap exclude even zero DM maxima.
573–589	PASS	All-size reverse-family positivity, common-domain inequalities and original residuals.
590–594	PASS	MB terminal cell and exact negative normalized maximum.
595–606	PASS	DM positive terminal mode with correct 37/256 subtraction bound and 91/256 margin.
607–616	PASS	DM concavity, strict terminal selection and continuation on the positive simple branch.
617–627	PASS	Rational seven-node tuple, own roots, strict positivity and original residuals.
628–638	PASS	Seventh-root cubic and all rational isolating signs independently verified.
639–654	PASS	Every MB representative negative; exact DM positive margin; strict selection of representative 2.
655–665	PASS	Separate full branch-labelled theorem with explicit lower branch and divisibility scope.
666–674	PASS	Full-domain 4-divides-n obstruction and strict cosine-grid rho bound in other sizes.
675–693	PASS	Lower-family coefficients, both exact positive roots, common domain and distinct branches.
694–710	PASS	Positive lower uniform value, convex selection, strict all-mode MB bound and rational six-node case.
711–718	PASS	Five existence categories reported without claiming each in every dimension.
719–738	PASS	n=8 two-unstable example: exact residuals, distinct modes 2/1, strict vertices and positive maxima.
739–745	PASS	Physical five-parameter openness at fixed size and named branch is stated precisely.
746–758	PASS	Correct upper-branch scalar IFT derivative, smooth MB root and finitely many strict gaps.
759–765	PASS	Correct lower-branch derivative and strict branch conditions; no uniform-in-n radius claimed.
766–779	PASS	Summary preserves all theorem quantifiers and permits open MB-stable regions.
780–788	PASS	Exact M=0 exponential and M=-1 blow-up witnesses; own DM roots; no MB positive root claimed.
789–799	PASS	Six-term, Hopf, transversality, criticality and later proof gates explicitly deferred.
800–806	PASS	Analytic checks and original-field residual identities are independently corroborated.

Lines	Status	Independent disposition
807–811	PASS	Located companion authenticated; loop counts independently derived and the original checker/results reproduced exactly.
812–814	PASS	Floating-point checks, interval certification, proof and independent audit are correctly distinguished.
815–846	PASS	Checklist and scope reminders; no new theorem or later-module premise is imported.

15 Closure disposition and deliverables

Required correction: E01 only. Correct the $n = 8$ universal witness's MB representative from 1 to 2, retaining the correct $n = 5, 6, 7$ statement. No mathematical theorem, parameter tuple, equilibrium, spectral formula, inequality, proof construction or branch condition needs correction. The located original validation companion is separately reproduced and creates no additional closure requirement. No other required change is identified.

The strongest single universal opposite-stability result independently verified here is that the same rational tuple $(1/4, 1/8, 1, 1, 1/8)$ has $x^* = y^* = 1$, unique uniform DM selection and $\mathrm{sa}(J_{\mathrm{DM}}) = -1/4$, but unique nonuniform MB selection and $\mathrm{sa}(J_{\mathrm{MB}}) \geq 3/16$ for every $n \geq 5, q = 2$. It persists on a nonempty physical open set for each fixed size. Equally substantial verified results are every ordered reachable pair with both models stable and the exact branch-qualified reverse-stability obstructions.

P07 clearance: P07 is cleared after the original development chat applies E01 to P08 and records audit closure for the corrected exact version. It is not begun or administratively cleared in this audit-only task. P08 remains DRAFT in all supplied project files. No later proof or full-project audited archive is created.

The audit-only archive is exactly `cyclic_network_project_A08_audit_only.zip`. Its sole project root is `cyclic_network_project/`; it includes this standalone source, its compiled PDF, the input/member hash inventory, the independent checker and outputs, the source-range ledger, and concise provenance and delivery checks. It excludes P08 source, all replacement project files, the original input archive, and build intermediates. The input ZIP retains its original name and bytes. Compilation and visual checks certify the reading copy, not mathematical truth.